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CircuitFlow: Smart Water Management, Drop by Drop

Urban Development, Mobility & Smart Cities: 73

Go, change the world

Team

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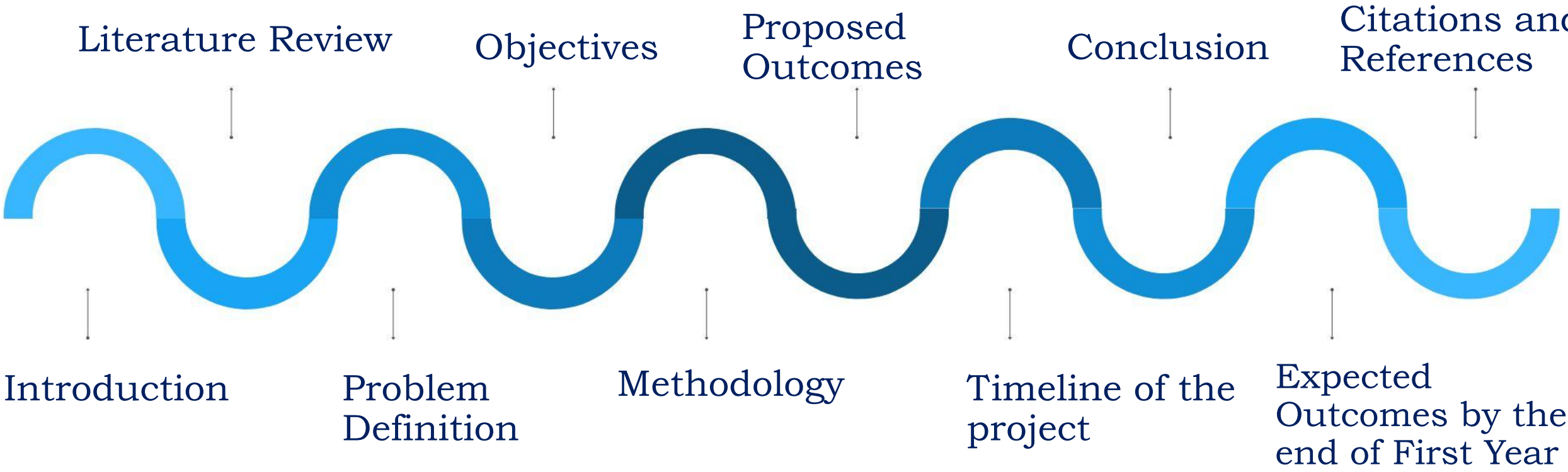
Suhas V J

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Outline

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Introduction

Overview

Smart tap-system for water wastage remediation and *usage analysis* at the household level through real-time data collection and *predictive modelling*.

Importance

- An average Indian household loses nearly 45,500 L through tap-leaks annually, aggravating a global water crisis.
- According to the India water portal report in Bangalore, around 360 million liters of water go waste every day due to leakage and other ‘minor’ reasons.
- This necessitates a structured wastage analysis and remediation system based on leakage-alerting, usage-dashboards, and data-driven regulation.

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Literature Review

CircuitFlow

Authors	Title of the publication	Foundations of the model	Plausible Upgrades/Fixes
Ruturaj Marathe, Manisha Tapale, Vidya Jadhav, Vrushab Hulbatte, Aishwarya Pawar	**IoT based Water Leakage Detection using Smart Objects for Smart City**	The model used Arduino flow-sensors to perceive leakages, and communicated the leakage data to the PushBullet application.	• Integration with solenoid valves to block leaks • analysis of usage-data and • timed water-flow mechanisms.
Chavhan S., Gawande U. H., Bhoyar D., Rangari S., Gardi M., & Gaikwad A.	**IoT-based Remote Control Water Distribution System (**International Journal of Intelligent Systems and Applications in Engineering)	• The model enables the control of water valves remotely through a mobile interface. • The system used a microcontroller(ESP8266), relay drivers, and solenoid valves to control water flow, aiming to reduce water losses and increase response time in saving water in large distribution pipelines.	• It relied on manual control rather than automatic shut off when water is left running. • It is aimed to be handled by the skilled personnel such as water operators, not individuals at the household level. • Integration of daily usage tracking and AI for analysing usage patterns.

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Literature Review

CircuitFlow

Authors	Title of the publication	Foundations of the model	Plausible Upgrades/Fixes
Vatsala Sharma, Kamal Nayanam, Himani	**Arduino based Smart Water Management**	• The paper lays an excellent foundation for IoT water management with the usage of sensors, though it focuses on automatic water irrigation. • The system uses Arduino (NodeMCU ESP8266), ultrasonic sensors to detect extent of water flow and Blynk app integration.	• Automatic shutoff which reduces wastage. • potential for AI-driven usage optimization • ***local data logging,*** making it not entirely reliant on stable internet connection.
Isuru Sachitha Herath	**IoT based Intelligent Domestic Water Management system**	• The paper proposes “Smart Water Buddy”, an IoT and machine learning-based system designed to monitor, analyze, and optimize domestic water usage.	• There is no failsafe mechanism to evade data loss/inconsistencies due to loss of internet connectivity.

Problem Definition

Problem Statement

- Rapid population growth and poor monitoring have intensified domestic water wastage, with Indian households losing about 45,500 L annually.
- Existing systems lack real-time control and user engagement.
- This project develops a smart, sensor-based system with flow-control valves and analytics to detect anomalies, automate corrections, and promote sustainable water use.

Problem Definition

Significance of the problem

- The increasing **global population** and **rapid urbanization** have intensified pressure on limited freshwater resources. Unmonitored household consumption and unnoticed tap leakages contribute significantly to domestic water wastage, particularly in countries like India.
- Addressing this issue is vital for ensuring sustainable water management and resource conservation. Implementing smart, real-time monitoring systems can **enhance user awareness,** reduce unnecessary losses, and promote responsible consumption, thereby contributing to long-term environmental sustainability and efficient water utilization in the residential sector.

Problem Definition

Problem context

- Water scarcity has become a critical global issue, with studies indicating that throughout 2025 nearly 67% of the world's population will experience moderate to high water stress.
- In India, urban households consume between 183 and 215 liters of water per person daily, placing immense pressure on already limited resources.
- The residential sector accounts for the largest portion of urban water usage, primarily for bathing, washing, flushing, and cooking.Bath and shower accounts for 55% of water-usage in urban households.

Objectives

Main Objective

- To effectively identify and address unattended water loss in the households of major Indian cities.

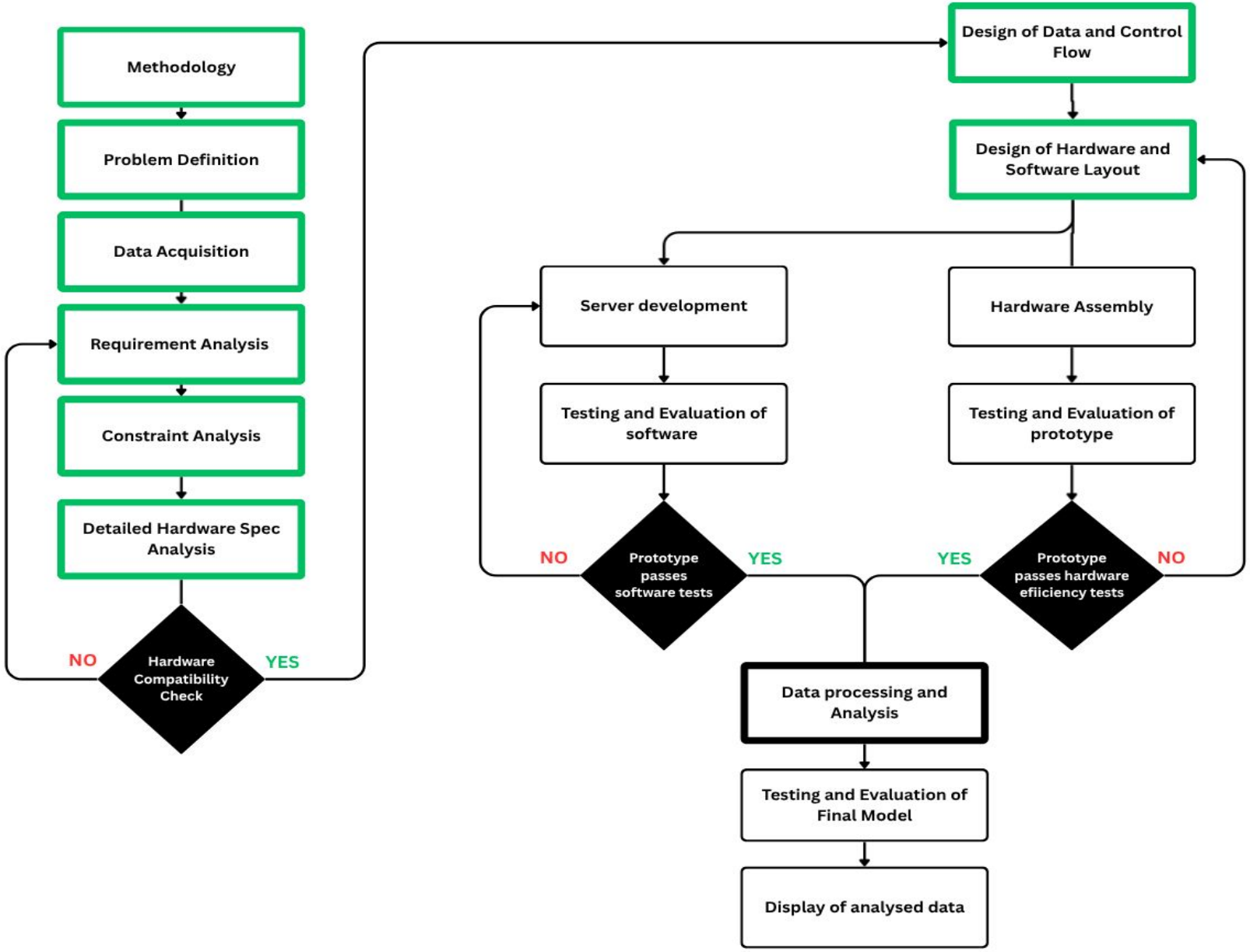
Specific Objectives

- To develop a model that detects and remediates leaking and overflowing taps.
- To demonstrate a dashboard on users' water utilization.

Expected Contributions to the field

- The model proposes to add a data-driven, adaptive approach to existing models on the subject.

Methodology



Methodology

STEP 1: Project Overview

This project adopts a *systematic approach*, detailed design, prototype development, and *performance evaluation*.

STEP 2: Data Acquisition:

Preliminary data was gathered from several *research papers* such as:

- Abdul Shaban, R N Sharma. Water Consumption Patterns in Domestic Households in Major Cities
- IoT-based Remote Control Water Distribution System
- Arduino based Smart Water Management.
- IoT-based Water Leakage Detection using Smart Objects for Smart Cities

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Methodology

STEP 3: Requirement Analysis and Problem Definition

The requirements were selected through careful analysis of *load conditions*, material availability, *power consumption*, safety standards, and an analysis of existing models.

STEP 4: Detailed Hardware Spec Analysis and Compatibility Checks:

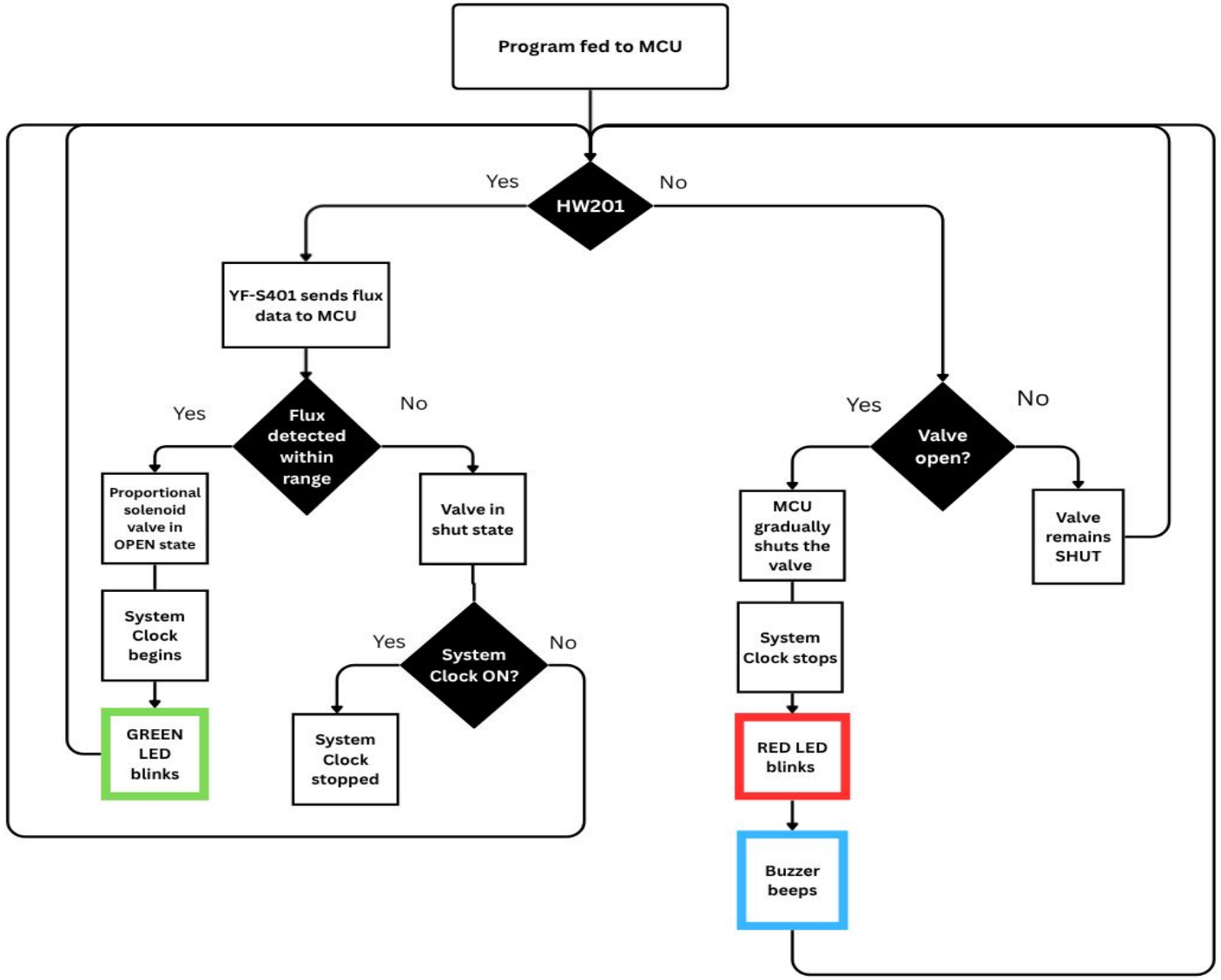
After performing detailed hardware spec analysis and compatibility checks, the following components were adopted for the model:

a. Arduino UNO MCU
b. Power supply and data extraction via USB-B
c. YF-S401 water-flow sensor
d. Standard jumper wires
e. Resistors
f. HW201 – IR obstacle avoidance sensor module
g. Proportional Solenoid valve .

STEP 5: Constraints

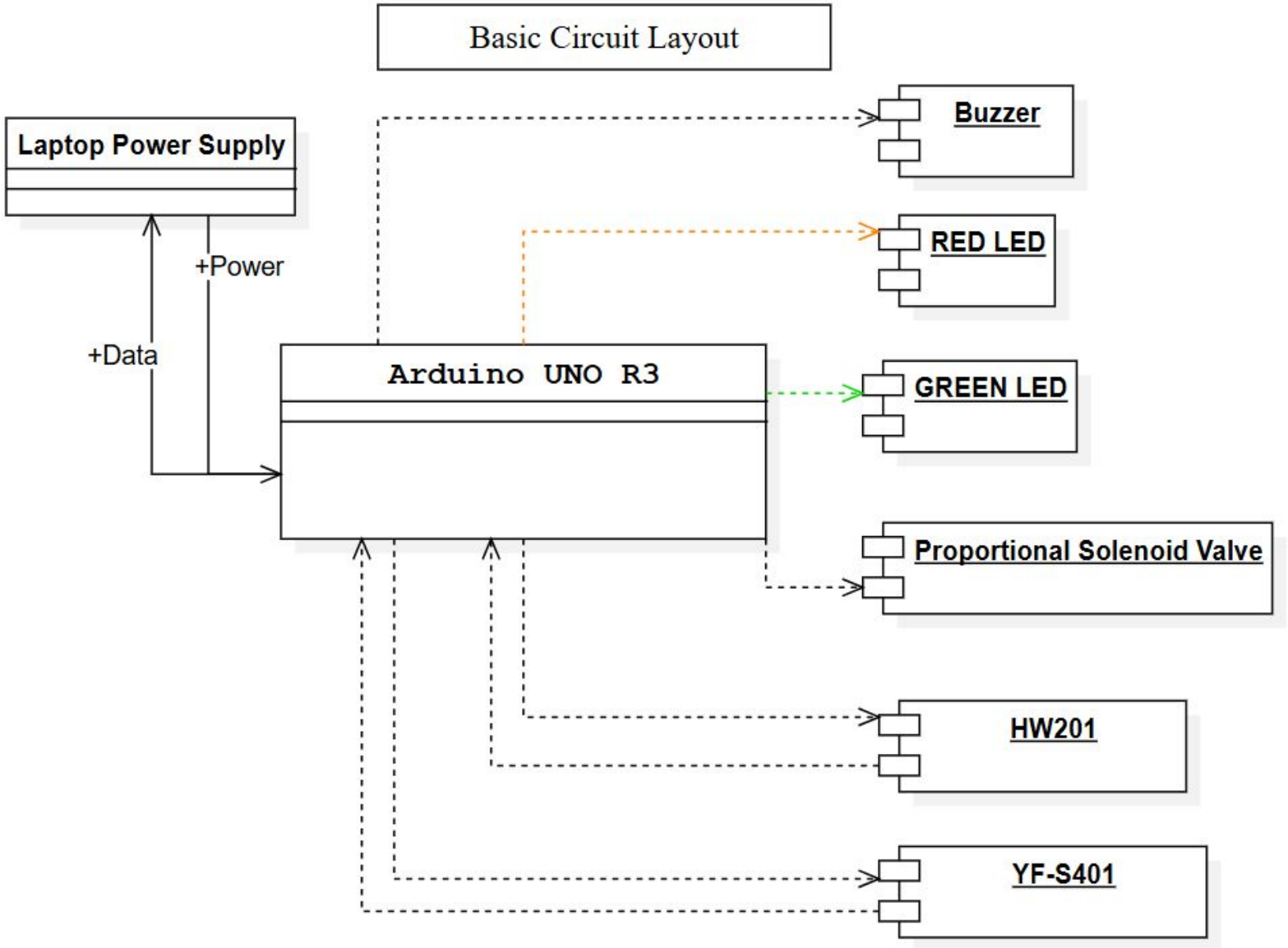
- For **maximum power efficiency**, the model can be operated in a specified timeframe , during which most indian households have maximum consumption.
- The model will operate only for a **maximum water flow pressure of 0.8MPa**.
- The model will operate on an **adjusted temperature range of 6-45℃** which is the usual water conditions.

STEP 6: Proposed Control Flow



The diagram showcases the proposed algorithmic strategy to be used by the model to *gauge data* from the user's consumption habits and regulate water loss.

STEP 7 : Proposed Hardware Layout



Methodology

STEP 8: Testing and Evaluation

A model reservoir and connecting pipes shall be fabricated. The prototype shall be mounted onto the apparatus.

1. Instrumental test: The hardware shall be tested for firm mechanical connections and electrical responsiveness.
2. Efficiency tests:
   a. Response-time
   b. Power efficiency(dB):

$$\rho = \log(p_1/p_2)$$

p_1= power output across the apparatus
p_2= power supplied to the apparatus

Methodology

STEP 9: Data Processing and Analysis

- The hardware model will collect data pertaining to *water flow rate* and *time of flow,* which will be sent to the laptop via USB-B cable.
- The model shall predict and estimate future use-habits based on the collected data.
- The collected data shall be formulated pictorially as a local dashboard, with early warnings for users.
- The collected data will be processed *locally* and made available for sharing via a *local server*.

Methodology

STEP 10: Ethical Consideration 

- There is significant concern on the security of users' data in existing models.
- Oftentimes, such models face lapses when there is no internet connection.
- This model shall process data and provide predictions *locally*, eliminating unsecured or unreliable internet connections and third-party applications.
- This also ensures that the data processing is faster and more reliable.

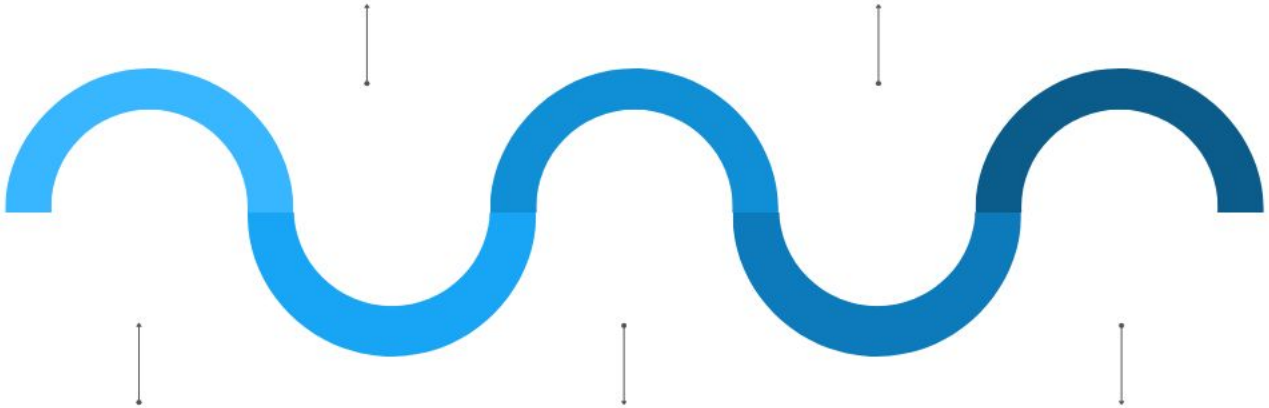
Timeline of the project

Week 1 – Review and Component Finalisation

- Finalize required hardware components (Arduino, sensors, valves, power setup) and their exact specifications.
- set up development environment (Arduino IDE, libraries).

Week 2 – Hardware Setup & Basic Programming

- Assemble the basic circuit: flow sensor, solenoid valve, relay, and controller.
- Write code for water flow detection and valve control.



Weeks 3–4 – System Integration & IoT Implementation

- Integrate IoT communication (ESP32 Wi-Fi / local server).
- Enable local data logging .
- Implement auto-shutoff logic for prolonged water flow.

Week 5 – Data Analysis & Efficiency Evaluation

- Collect and analyze usage data to measure water-saving efficiency.
- Apply simple mathematical formulations for flow rate and wastage reduction.
- Document observations and identify potential improvements.

Week 6 – Final Testing, Documentation & Presentation

- Final prototype demonstration (hardware + software).
- Prepare final documentation, charts, and results for review.
- Noting findings, challenges, and future upgrade ideas

Proposed Outcomes

Expected Results

- Systematic analysis of water-consumption data of households in major Indian cities.

Potential Impact

- For every liter of water saved every day, the annual savings aggregate to over 1000 olympic pools.
- This model could significantly assist in combating the existing water-exigency.

Future Research

This project enables research in IoT-based systems, training of machine-learning models based on household water consumption.

Conclusion

Summary:

By integrating low-power sensors, a flow-control valve(actuator), and a local real-time analytical dashboard, the system aims to promote sustainable usage behaviour.

Further Work:

Further extensions to the proposed model include mobile-based monitoring, and real-time regulations for extended time frames.

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Expected Outcome by the end of First Year

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Research Proposal

This project holds potential for research-extension.

Expected Budget (preliminary estimate)

Components	Estimated Cost(in INR)
• Arduino UNO R3	• 1400-1700
• YF-S401	• 200
• HW-201	• 40
• Proportional solenoid valve	• 3500-4000
• Wirings+Resistors	• 350-500
• LEDs	• 30-40
• Buzzer	• 40

References

List of References:

- NeeRain: 49 billion liters of water is wasted daily in the country due to carelessness - NeeRain
- BelTechno Global:Water Leakage, A Big Issue In Water Distribution And Storage System In India.
- Abdul Shaban, R N Sharma. Water Consumption Patterns in Domestic Households in Major Cities
- IEEE Xplore

Citations

- Ruturaj M et al. (2021). IoT based Water Leakage Detection using Smart Objects for Smart City
- Smart System for Stabilizing Water Flow Output on Android- Based Taps. Iwan Suhardi et al.(2023).
- Chavhan, S. , Gawande, U.H. , Bhoyar, D. , Rangari, S. , Gardi, M. and Gaikwad, A. (2023). IoT-based Remote Control Water Distribution System. International Journal of Intelligent Systems and Applications in Engineering. 12, 7s (Dec. 2023), 572–583.
- Vatsala Sharma , Kamal Nayanam , Himani, 2020, Arduino based Smart Water Management, INTERNATIONAL JOURNAL OF ENGINEERING RESEARCH & TECHNOLOGY (IJERT) Volume 09, Issue 08 (August 2020)